Mines Paris PSL - Etilab

Pooling Contracts between Venture Capital Institutions

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Outline

Contracts between VC institutions

- **VCs $\leftrightarrow$ Limited Partners.**
  - **Ex-ante commitment.** In 93% of deals, VC invest in a portfolio of companies and LPs commit capital before knowing any target company *(Lerner et al., 2022)*.
  - **Pooling investment split.** The investment split is fixed at 1-5% of the total cost of investment.
  - **Pooling return split.** The return split is fixed at 20% of the return plus 2% of committed capital.
- **Accelerators $\leftrightarrow$ Follow-on VCs.**
  - **Ex-post commitment.** YC and Techstars organize a demo day to present their batches of startups to potential follow-on VCs before they engage.
  - **Pooling investment split.** Before 2022, the investment amount is fixed for each startup at $125K for YC and $120K for Techstars.
  - **Pooling return split.** Before 2022, the return split was fixed at 7% for YC and 6% for Techstars

Two relationships, two timings

Accelerator $\longrightarrow$ Follow-on VCs

Accelerator sets batch terms $\rightarrow$ **Demo Day:** π revealed $\rightarrow$ VCs commit deal by deal $\rightarrow$ Execute

investors learn π *before* committing $\Rightarrow$ **ex-post commitment**

VCs $\longrightarrow$ Limited Partners

VC proposes partnership terms $\rightarrow$ LPs commit capital $\rightarrow$ **Capital call:** π revealed $\rightarrow$ Execute

investors commit *before* learning π $\Rightarrow$ **ex-ante commitment**

Literature Review

- **Performance of contracts between VCs and LPs.**
  - Lerner and Nanda (2020); Metrick and Yasuda (2010): Management fees incentivize funds to grow at the expense of performance.
  - Axelson et al. (2009); Fang et al. (2015): Blind pool contracts minimizes VCs' incentives to overstate projects' values.
  - Gourier et al. (2024); Lerner and Nanda (2020): LP's demand for VC is high (and concentrated).
- **Performance of the accelerator model.**
  - Gonzalez-Uribe and Leatherbee (2017): Increases VC fundraising and scale.
- **Theory of contracts.**
  - Myerson (1983): A common mechanism can be proposed by an informed principal whatever the specific projects encountered.
  - Maskin and Tirole (1992); Myerson (1983); Holmstrom and Myerson (1983); Guesnerie and Laffont, (1984); Kamien and Schwartz, (1991)

This paper

A first analysis of contracts between venture capital institutions, in an informed-principal framework with competitive agents.

Striking findings

- Under both regimes, pooling contracts are strategically optimal for the agency.
- Under ex-post commitment:
  - the unique optimal contract is pooling,
  - the agency's budget is always saturated,
  - some good projects always remain unfunded.

The agents

Entrepreneurs

A project needs a sunk cost $I \in (0, 1)$, yields an outcome in $\{0, 1\}$ with success probability π. Density f, c.d.f. F. No effort cost: the entrepreneur needs no equity.

The agency

- Budget B_a
- Knows π for each project
- Profit-maximizing

Investors

- Budget B_i
- Know only the distribution of π
- Competitive (price-taking)

Allocation $\mu = (\sigma, \rho)$: the agency contributes $\sigma(\pi)$ to the cost I and keeps a share $\rho(\pi)$ of the outcome; the investor completes (and takes) the rest.

Payoffs and assumptions

Payoffs. For each project π:

$$U_a(\pi) = \rho(\pi)\pi - \sigma(\pi) \qquad U_i(\pi) = \big(1 - \rho(\pi)\big)\pi - \big(I - \sigma(\pi)\big)$$

Budget constraints. $B_a \geq \int \sigma(\pi) f(\pi)\, d\pi$ and $B_i \geq \int (I - \sigma(\pi)) f(\pi)\, d\pi$

Assumptions

- **(A1)** $\mathbb{E}(\pi) - I < 0 < 1 - I$, and f log-concave: the average project is not worth financing, the best one is.
- **(A2)** $B_a < I(1 - F(I)) \leq B_a + B_i$: the agency cannot finance all good projects alone, but *together* they can.
- **(A3)** The agency is profit-maximizing.

Game 1 (Benchmark)

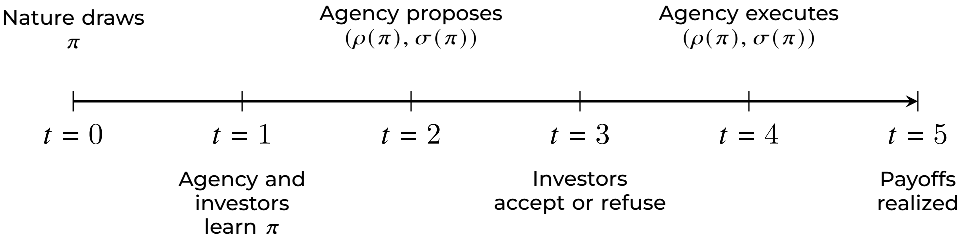


Figure: Game 1 – Benchmark

Game 1 (Benchmark)

Proposition 1

Under budget and investors participation constraints:

The agency maximizes profit $\Leftrightarrow$ Investor utility $U_i(\pi) = 0$ for each π

The agency finances all good projects

Investors participation constraint: $U_i(\pi) \geq 0$ for each π

Game 1 (Benchmark)

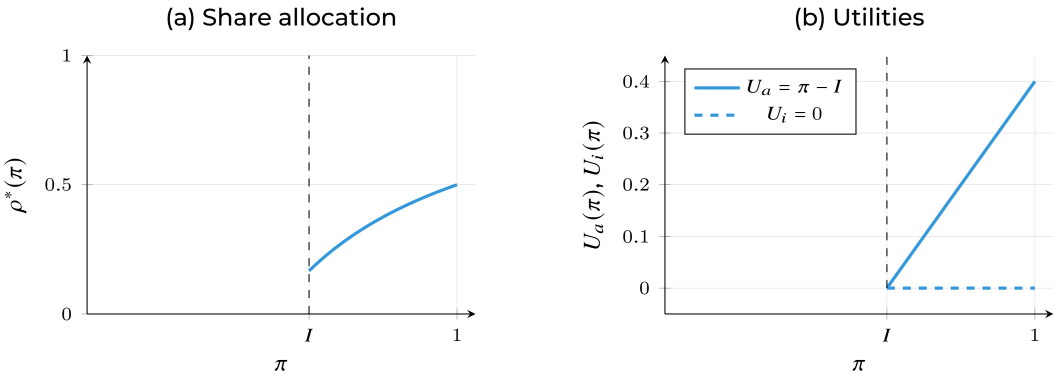


Figure: Parameters: $\pi \sim U[0,1]$, $I = 0.6$, $\sigma^* = 0.1$. (a) The share $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ is increasing in π. (b) The agency captures the full surplus $\pi - I$; the investor earns zero rent.

Game 2 (Ex-post commitment)

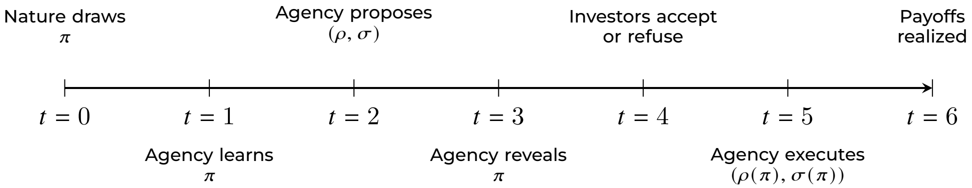


Figure: Game 2 – Ex-post commitment

Game 2 (Ex-post commitment)

Proposition 2

Under budget and investors participation constraints:

The agency maximizes profit $\Leftrightarrow$

- The allocation is pooling
- The agency saturates its budget
- $U_i(\pi_{\min}) = 0$ for the less good financed project
- The profit-maximization FOC is satisfied

Investors participation constraints:

- $U_i(\pi) \geq 0$ for each π
- $U_a(\pi) \geq \rho(\hat{\pi})\pi - \sigma(\hat{\pi})$ for each $(\pi, \hat{\pi})$

Game 2 (Ex-post commitment)

Remark

$\pi_{\min} > I$: Some good projects always remain unfunded.

Game 2 (Ex-post commitment)

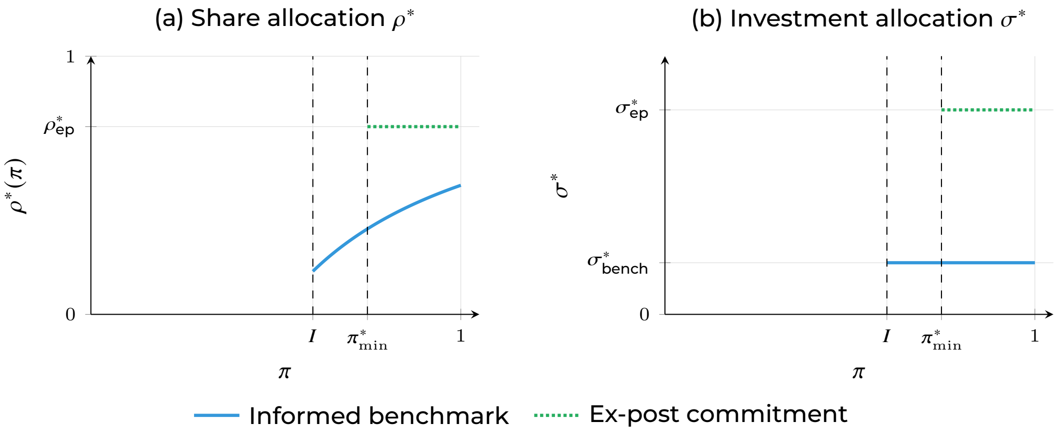


Figure: Informed benchmark vs. ex-post commitment regime. Parameters: $\pi \sim U[0, 1]$, $I = 0.6$, $B_a = 0.1$, $B_i = 0.2$.

Game 2 (Ex-post commitment)

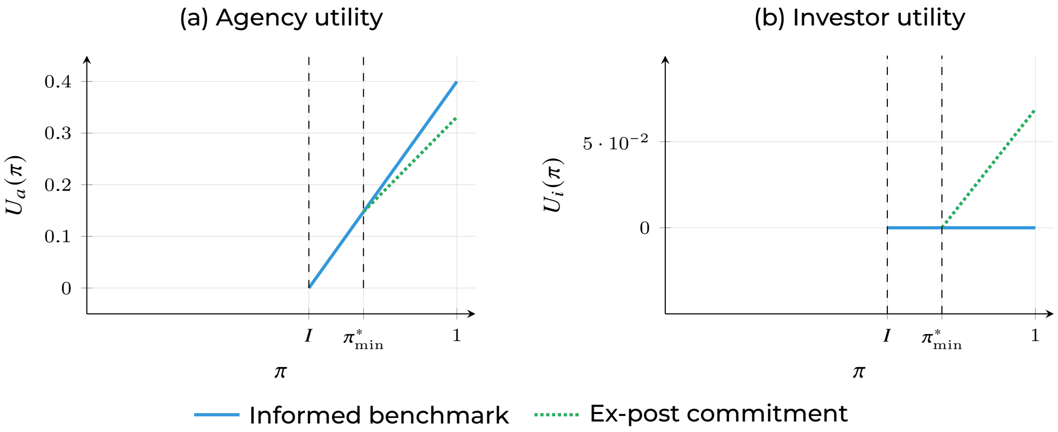


Figure: Utilities per project: informed benchmark vs. ex-post commitment regime. Same parameters.

Game 3 (Ex-ante commitment)

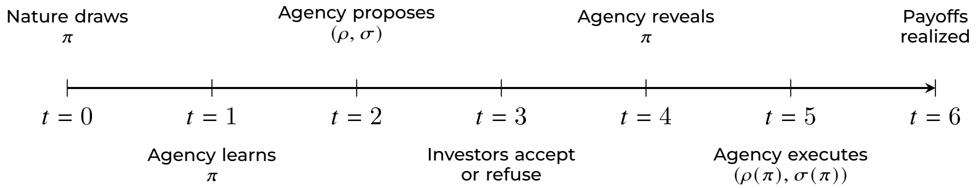


Figure: Game 3 – Ex-ante commitment

Game 3 (Ex-ante commitment)

Proposition 3

Under budget and investors participation constraints:

The agency maximizes profit	$\Longleftarrow$	The allocation is pooling The agency does not saturate its budget Investor average utility $\int U_i(\pi) f(\pi) d\pi = 0$ The agency finances all good projects

Investors participation constraint: $\int U_i(\pi) f(\pi) d\pi \geq 0$

Game 3 (Ex-ante commitment)

Remark

The reverse is not true.

Game 3 (Ex-ante commitment)

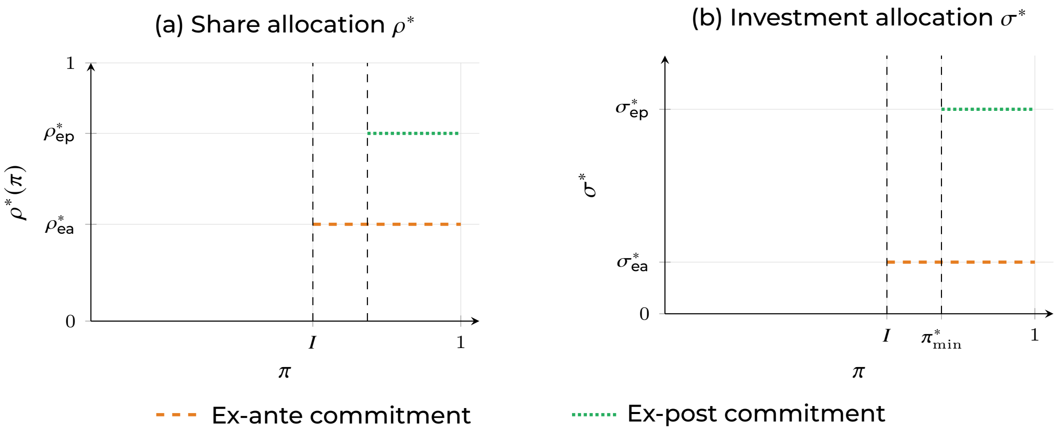


Figure: Optimal allocation under ex-ante vs. ex-post commitment investors. Same parameters as in Game 2.

Game 3 (Ex-ante commitment)

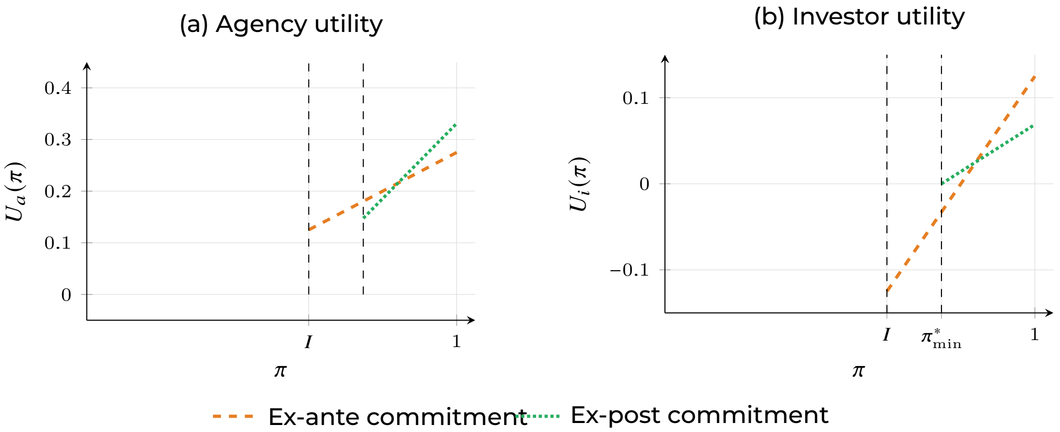


Figure: Utilities under the two commitment regimes. Same parameters as in Game 2.

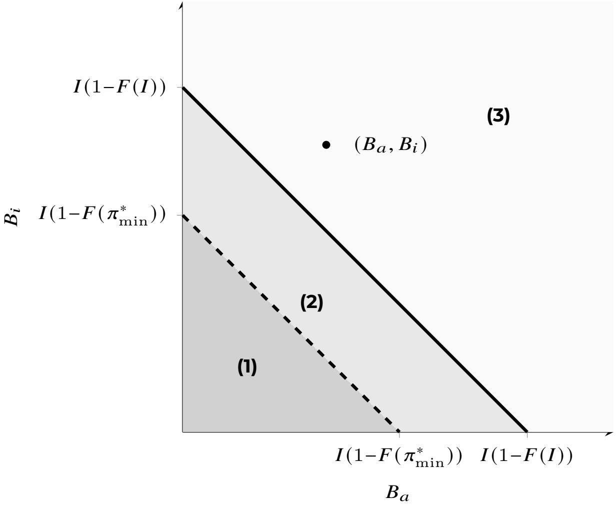
Cash-poor vs. cash-rich economies

We restrict to pooling contracts and we relax (A2): the aggregate budget may not be enough to finance all good projects.

Definitions: Severely cash-poor, moderately cash-poor, cash-rich economies

- **Severely cash-poor economy.** Aggregate budget not enough to finance projects above the ex-post threshold
- **Moderately cash-poor economy.** Aggregate budget enough to finance projects above the ex-post threshold but not enough to finance all projects
- **Cash-rich economy.** Aggregate budget enough to finance all projects

Cash-poor vs. cash-rich economies



(1) Severely cash-poor: both regimes coincide; only the budget binds
(2) Moderately cash-poor: ex-ante efficient, ex-post inefficient
(3) Cash-rich: ex-ante efficient, ex-post inefficient

Takeaways

Main result

- Under ex-ante commitment, there is an optimal efficient pooling contract but pooling is unnecessary;
- Under ex-post commitment, the optimal contract is uniquely pooling, budget-saturating but inefficient.

- **Pooling contracts rationalized.** A common, pooling mechanism is optimal for contracts between institutions in VC.
- **The discipline device.** A binding agency budget constraint gives the agency skin in the game and makes investors confident it will not misvalue projects — at the cost of financing fewer of them.
- **Institutions, mapped.** Ex-post commitment $\leftrightarrow$ accelerator–VC (Demo Day); ex-ante commitment $\leftrightarrow$ VC–LP (capital call).
- **Policy.** In cash-poor economies, the accelerator model is efficient. In cash-rish economies, it leaves some good projects unfunded.

Thank you

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